



SK-DGCNN: Human activity recognition from point cloud data with skeleton transformation

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ABSTRACT

Human Activity Recognition (HAR) has become a prominent research topic in artificial intelligence, with applications in surveillance, healthcare, and human–computer interaction. Among various data modalities used for HAR, skeleton and point cloud data offer strong potential due to their privacy-preserving and environment-agnostic properties. However, point cloud-based HAR faces challenges like data sparsity, high computation cost, and a lack of large annotated datasets. In this paper, we propose a novel two-stage framework that first transforms radar-based point cloud data into skeleton data using a Skeletal Dynamic Graph Convolutional Neural Network (SK-DGCNN), and then classifies the estimated skeletons using an efficient Spatial Temporal Graph Convolutional Network++ (ST-GCN++). The SK-DGCNN leverages dynamic edge convolution, attention mechanisms, and a custom loss function that combines Mean Square Error and Kullback–Leibler divergence to preserve the structural integrity of the human pose. Our pipeline achieves state-of-the-art performance on the MMActivity and DGUHA datasets, with Top-1 accuracy of 99.73% and 99.25%, and F1-scores of 99.62% and 99.25%, respectively. The proposed method provides an effective, lightweight, and privacy-conscious solution for real-world HAR applications using radar point cloud data.

1. Introduction

Human Activity Recognition (HAR) is a core area of research in computer vision and artificial intelligence, enabling machines to perceive, interpret, and respond to human behaviors. Its applications span various real-world domains, including healthcare monitoring, smart surveillance, human–computer interaction, and assisted living (Dinarević et al., 2019; Hamad et al., 2022; Karim et al., 2024). It is important to design robust and scalable HAR systems capable of operating in complex environments such as low-light conditions. The Internet of Things (IoT), such as cameras, radars, and wearable sensors, has advanced the development and deployment of real-life applications (Fortino et al., 2014; Mukhopadhyay, 2014; Zhou et al., 2020). Traditional HAR methods have predominantly relied on RGB or RGB-D video data with depth information. With the availability of large-scale public datasets (Caba Heilbron et al., 2015; Kay et al., 2017; Reiss & Stricker, 2012; Shahroudy et al., 2016; Soomro et al., 2012), researchers have applied deep learning techniques for activity classification using 2D or 3D visual cues. While these approaches have achieved high accuracy under controlled conditions, their real-world deployment remains limited due to several challenges, including sensitivity to environmental conditions such as lighting or occlusion, high computational demands for real-time analysis, and critical privacy

concerns when capturing personally identifiable video data (Ke et al., 2013). Wearable sensors offer a more private and structured way to capture human motion. However, these devices are often intrusive, require user compliance, and may lead to discomfort or inconsistent data quality due to improper usage (Mukhopadhyay, 2014; Zhou et al., 2020). These limitations make video and wearable-based HAR unsuitable for long-term, passive monitoring in applications such as elder care, patient tracking, or occupancy monitoring in private settings.

Recently, radar-based point cloud data has emerged as a promising alternative. mmWave radars generate sparse point cloud representations of human movement using frequency-modulated continuous wave (FMCW) signals (An & Ogras, 2021; Singh et al., 2019). These sensors are cost-effective, privacy-preserving, and robust to lighting and background changes. They have proven to be particularly effective for indoor applications such as fall detection, gesture recognition, and activity monitoring under challenging visibility conditions (Ullmann et al., 2023). However, point cloud data poses unique challenges: it is often sparse and unstructured, lacks semantic context, and is not directly applicable to traditional deep learning pipelines without significant preprocessing.

Existing work in the literature explored direct classification of point cloud data using voxelization (Zhou & Tuzel, 2018) or handcrafted

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2D histogram representations (Luo et al., 2023). While effective, these methods are computationally intensive and often fail to capture fine-grained motion features necessary for differentiating similar actions. The scarcity of annotated radar point cloud datasets further limits the use of supervised learning in this domain. In contrast, skeleton-based HAR (SB-HAR) methods have demonstrated excellent performance due to their ability to model human motion using graph structures where joints are treated as nodes and limbs as edges (Duan et al., 2022; Yan et al., 2018). These methods are lightweight, interpretable, and well-suited to GCN-based architectures. Leveraging this, recent research has proposed converting radar point clouds into skeleton keypoints (Lee & Kim, 2022), allowing existing SB-HAR pipelines to be reused with radar data. This strategy provides the best of both modalities, privacy-aware sensing, and efficient modeling techniques.

Despite these advances, existing radar-to-skeleton approaches face critical limitations. First, CNN-based pose estimators (An & Ogras, 2021; Lee & Kim, 2022) treat point clouds as fixed-size inputs and fail to capture dynamic local geometric relationships, leading to reduced accuracy in limb extremities. Second, existing methods optimize only for pointwise coordinate accuracy using MSE loss, neglecting the preservation of global skeletal structure and anatomical constraints, which are essential for distinguishing similar activities. Third, the training strategies employed do not account for the varying difficulty of estimating coordinates along different spatial axes, resulting in suboptimal convergence. These gaps motivate the need for a graph-based pose estimation model with structure-aware loss functions and axis-specific training strategies.

In this paper, we propose a novel two-stage framework to bridge the gap between radar point cloud data and skeleton-based activity recognition. Specifically, we introduce a Skeletal Dynamic Graph Convolutional Neural Network (SK-DGCNN) that estimates 3D human keypoints from radar-based point clouds. Our model builds upon DGCNN (Wang et al., 2019), enhanced with an Efficient Channel Attention (ECA) module (Wang et al., 2020) and a new loss function that combines Mean Squared Error (MSE) with Kullback–Leibler (KL) Divergence (KLDiv) to preserve both joint-level precision and the global pose shape. In the second stage, we apply the lightweight yet powerful ST-GCN++ model (Duan et al., 2022) to perform action classification using the estimated skeleton data. We validate our framework on two publicly available datasets: MMActivity (Singh et al., 2019) and DGUHA (Lee & Kim, 2023), achieving strong and reproducible performance with mean Top-1 accuracies of $99.59 \pm 0.22\%$ and $98.72 \pm 0.33\%$, respectively, across three independent training runs per dataset. Our pipeline eliminates the need for complex voxelization, reduces computational cost, and enables the deployment of HAR systems in privacy-sensitive and resource-constrained settings.

To summarize, our contributions are as follows:

- We propose **SK-DGCNN**, a dynamic graph CNN that transforms sparse radar point cloud data into accurate 3D skeleton keypoints using edge-based feature learning with an efficient attention mechanism.
- We introduce a **novel shape-preserving loss function** that combines MSE with KL divergence to enforce both joint-level precision and global pose structure preservation. Unlike conventional regression losses that optimize coordinates independently, our loss treats the skeleton as a spatial distribution, ensuring anatomical plausibility and improving downstream classification performance even when pointwise errors are similar.
- We propose a **sequential multitask learning strategy** that trains coordinate axes progressively ($z \rightarrow x \rightarrow y$), accounting for the varying difficulty of spatial estimation and improving convergence stability.
- We integrate SK-DGCNN with the ST-GCN++ model to build a complete end-to-end HAR pipeline, benefiting from structured skeleton data and efficient spatio-temporal reasoning.
- We achieve competitive accuracy and F1 scores on the MMActivity and DGUHA datasets, validating the effectiveness and generalizability of our approach.

2. Related work

2.1. Radar point cloud data for HAR

Millimeter wave (mmWave) radar sensors have become increasingly popular for HAR due to their robustness to lighting conditions and their ability to preserve user privacy. Early works such as Singh et al. (2019) introduced the MMActivity dataset and used voxelization followed by CNN and Bi-LSTM models. However, voxelization increases memory usage and leads to sparse, inefficient representations (Deng et al., 2021; Zhou & Tuzel, 2018). To improve upon this, An and Ogras (2021) proposed the MARS dataset and a CNN-based pose estimation model. Recent advances by Lee and Kim (2023) extended this with the DGUHA dataset, combining radar and skeleton data for cross-modal learning. More recently, Luo et al. (2023) introduced EdgeActNet, which converts point clouds into 2D histograms for efficient classification. Li et al. (2023) proposed mmBehavior, a deep recurrent neural network approach for radar point cloud activity recognition. Ullmann et al. (2023) provided a comprehensive survey of radar-based continuous HAR, highlighting ongoing challenges in generalization and real-world deployment. Other pipelines using CNNs and LSTMs have also been proposed (Anwar, 2022; Jin et al., 2019; Yu et al., 2022).

While effective, these approaches often depend on handcrafted pre-processing or are constrained by limited annotated datasets. Our work addresses these challenges by learning to convert point clouds into skeleton representations, enabling the use of established graph-based HAR techniques.

2.2. Point cloud to skeleton estimation

To bridge the gap between sparse radar point clouds and structured representations of human motion, recent studies have explored converting point cloud data into skeleton keypoints. Lee and Kim (2022) proposed the use of a CNN model to estimate skeletons from radar point clouds and then applied ST-GCN for HAR. While effective, their CNN-based pose estimation approach showed limited precision in limb extremities, which are crucial for distinguishing subtle actions. Wang et al. (2020a) and Sengupta and Cao (2022) explored direct regression of keypoints from point cloud data using convolutional models, but their evaluations show challenges in generalizing across diverse movements and occlusions. Recent surveys on human pose estimation (Dubey & Dixit, 2023; Zheng et al., 2023) highlight the continued challenges in 3D pose estimation from sparse and noisy sensor data, particularly in real-world deployment scenarios.

Our approach differs by using a graph-based model (SK-DGCNN) that dynamically learns edge relationships from raw point clouds, capturing both local and global spatial features for accurate skeleton estimation. This structure-aware learning is particularly suitable for human pose modeling.

2.3. Graph models for skeleton-based HAR

Skeleton-based HAR (SB-HAR) has become a dominant paradigm due to its structured, compact, and semantically rich representation of human motion. Yan et al. (2018) introduced ST-GCN, which models both spatial and temporal dependencies in skeleton data using a graph convolutional network. Subsequent improvements, such as attention mechanisms (Li et al., 2019), multi-stream architectures (Shi et al., 2019), and disentangled representations (Anand et al., 2026; Liu et al., 2020; Shi et al., 2020) further improved performance and generalization of SB-HAR. Duan et al. proposed ST-GCN++ (Duan et al., 2022), a more efficient and modular GCN that replaces monolithic temporal convolutions with multi-branch TCNs, and enhances spatial modeling through adaptive edge updates. Recent work has explored transformer-based architectures (Do & Kim, 2024; Teng et al., 2025; Zhou et al.,

3D Pose Estimation Converting from Point Cloud

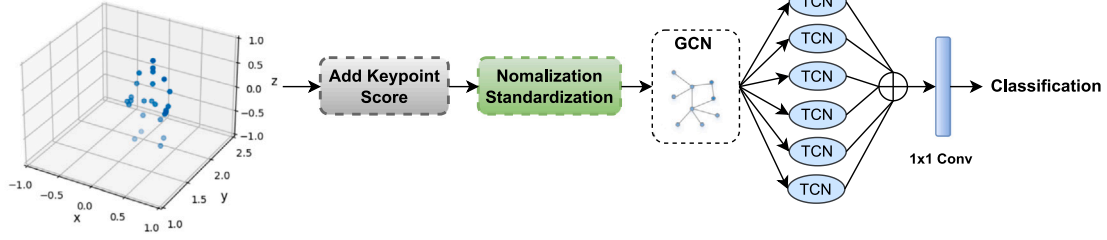


Fig. 1. End-to-end pipeline for skeleton-based HAR using the transformed skeleton data from the radar point cloud data.

2022) and topology-aware convolutions (Xu et al., 2022) for skeleton-based action recognition, achieving strong results on large-scale benchmarks. Additionally, (Eskandari et al., 2026) demonstrated that prioritizing structurally influential nodes improves graph-based learning efficiency. These models have consistently outperformed CNN- and RNN-based baselines on benchmark datasets like NTU-RGB+D (Shahroudy et al., 2016) and Kinetics-400 (Kay et al., 2017).

By transforming radar point clouds into skeleton data using SK-DGCNN, we enable the application of ST-GCN++ to radar-based HAR. This integration benefits from the best properties of both domains: robust sensing and graph-based temporal modeling.

2.4. Point-cloud models for unstructured data

Processing unstructured point cloud data requires models that can capture local geometric structures without imposing grid-like assumptions. PointNet (Qi et al., 2017) was an early attempt, but it treated points independently and failed to model local interactions. Dynamic Graph CNN (DGCNN) (Wang et al., 2019) addressed this by constructing a local neighborhood graph using K-nearest neighbors and learning edge features dynamically through EdgeConv. While DGCNN has been widely adopted in object recognition and segmentation, its use in HAR and human pose estimation is limited. Our proposed SK-DGCNN extends DGCNN by integrating Efficient Channel Attention (ECA) (Wang et al., 2020) and a multitask loss function to estimate 3D human skeletons from sparse radar point clouds, which are then passed to graph-based HAR models. This approach eliminates the need for voxelization and maintains structural consistency in pose representation, improving downstream classification accuracy.

3. Methodology

We propose a two-stage framework for Human Activity Recognition (HAR) using radar point cloud data. The first stage converts sparse point cloud data into 3D human skeletons, and the second stage uses the skeleton data for action classification. Fig. 1 illustrates the complete pipeline of our proposed method.

3.1. Stage 1: Radar point cloud to skeleton

Radar point cloud data is sparse and unstructured, making it unsuitable for conventional deep learning techniques. To address this, we propose SK-DGCNN, an enhanced version of Dynamic Graph Convolutional Neural Network (DGCNN) (Wang et al., 2019), which directly estimates 3D skeleton joint coordinates from point clouds.

3.1.1. Graph construction and edge feature learning

Given a set of points $P = \{p_1, p_2, \dots, p_n\}$ in a 3D radar frame, SK-DGCNN constructs a dynamic graph using the k -nearest neighbors (K-NN) and learned edge features via EdgeConv. We set $k = 20$ to balance local geometric coverage and computational efficiency. With radar point clouds containing approximately 25 points per frame, $k =$

20 captured roughly 80% of the local neighborhood, ensuring sufficient context for edge feature learning while maintaining manageable computational cost. This choice is consistent with the original DGCNN work (Wang et al., 2019) and provides an effective receptive field for modeling human pose structure from sparse radar data. The edge feature between two points p_i and p_j was computed as:

$$e_{ij} = h_{\theta}(p_i, p_j) = \theta_m \cdot (p_i - p_j) + \phi_m \cdot (p_i). \quad (1)$$

In Eq. (1), e_{ij} denotes the learned edge feature between the central point p_i and its neighbor p_j . Here, $p_i, p_j \in \mathbb{R}^3$ represent 3D point coordinates in a radar frame, h_{θ} is the edge feature function parameterized by learnable weights, θ_m and ϕ_m . The first term $\theta_m \cdot (p_i - p_j)$ captures the relative spatial difference between the two points (local geometry), while the second term $\phi_m \cdot p_i$ incorporates the absolute position of p_i (global structure). M denotes the number of convolutional filters or channels in the EdgeConv layer. The EdgeConv modules extracted multi-level features, which were then aggregated and passed through 1D convolutions and MLP layers to regress the x , y , and z coordinates of skeleton joints (see Fig. 2).

3.1.2. Efficient channel attention (ECA)

To improve feature learning without increasing computational cost, we used the Efficient Channel Attention (ECA) module (Wang et al., 2020). The kernel size of the 1D convolution is computed as:

$$k = \left\lceil \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right\rceil_{\text{odd}}, \quad (2)$$

where C is the number of channels, and $\gamma = 2$, $b = 1$ are hyperparameters selected based on empirical performance.

3.1.3. Loss function

While MSE loss is effective in minimizing joint-level errors, it does not enforce structural consistency. To capture both precision and shape similarity, we combined MSE with KLDiv loss as shown in Eqs. (3) and (4):

$$L_c = \alpha_{MSE} L_{MSE_c} + \alpha_{KLDiv} L_{KLDiv_c} \\ = \alpha_{MSE} \frac{\sum_{i=1}^n (Pred_{c_i} - GT_{c_i})^2}{n} + \alpha_{KLDiv} \frac{\sum_{i=1}^n (GT_{c_i} \cdot \log(\frac{GT_{c_i}}{Pred_{c_i}}))}{n} \quad (3)$$

where α denotes the weighting coefficient for each loss term, and the subscript c indicates the spatial coordinate axis (x , y , or z). The total loss across all coordinates is computed as:

$$L_{\text{total}} = L_x + L_y + L_z \quad (4)$$

We used dynamic weighting as shown in Eq. (5) to shift the focus from MSE to KLDiv gradually over training epochs:

$$\alpha_{MSE} = 1 - 0.05 \times \left\lceil \frac{n_{\text{Epoch}}}{20} \right\rceil \\ \alpha_{KLDiv} = 0.05 \times \left\lceil \frac{n_{\text{Epoch}}}{20} \right\rceil \quad (5)$$

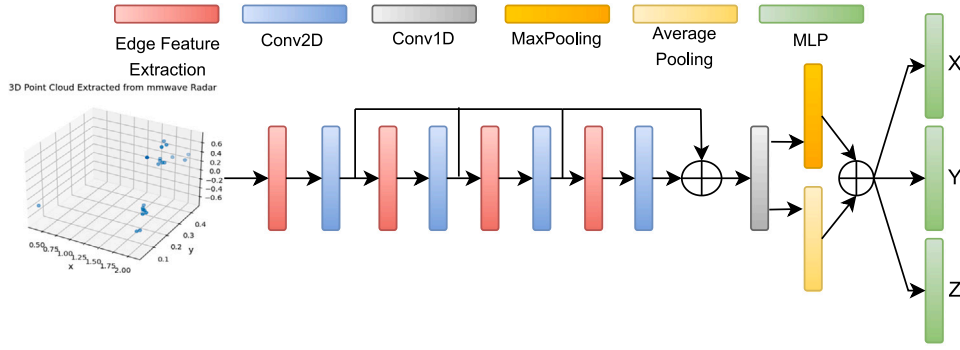


Fig. 2. SK-DGCNN for skeleton estimation.

The rationale behind this dynamic weighting schedule is that it helps in balancing early-stage coordinate learning with late-stage structure preservation. In early epochs, the model benefits from strong MSE supervision ($\alpha_{MSE} = 1.0$) to learn accurate joint positions from scratch. As training progresses, the KLDiv weight (by 0.05 every 20 epochs) can be gradually increased to enforce distributional similarity between predicted and ground truth skeletons, ensuring that the model preserves anatomical structure. This curriculum-style approach prevents the KL-Div term from destabilizing early training while allowing it to refine pose geometry in later stages. The step size of 0.05 and interval of 20 epochs were selected empirically to provide smooth transitions without abrupt loss landscape changes.

3.1.4. Sequential multitask learning

To improve convergence, especially along challenging axes, we employed sequential multitask learning (Sun et al., 2020). The model was first trained to regress the vertical (z) coordinates, followed by the horizontal (x) and depth (y) coordinates in sequence. This progressive learning strategy reduced the chance of forgetting and helped stabilize training. The ordering ($z \rightarrow x \rightarrow y$) was motivated by both empirical observation and physical intuition. The vertical axis (z) exhibits the most consistent signal in radar data due to the sensor's elevation angle and the prominence of torso movements, making it the easiest to learn, and provides a stable foundation. The horizontal axis (x) captures lateral motion and is moderately challenging. The depth axis (y) is the most difficult due to radar range resolution limitations and occlusion effects. Training in order of increasing difficulty allows the model to build robust feature representations progressively, with earlier tasks providing useful inductive biases for the later ones. As shown in Fig. 7, this ordering yielded lower validation losses compared to alternative sequences.

3.2. Stage 2: Skeleton-based action classification

Once the skeleton data was estimated, we applied a graph convolutional model to classify human actions. We use ST-GCN++ (Duan et al., 2022), a high-performance, efficient framework that models spatial relationships using GCNs and captures temporal dynamics, with a multi-branch Temporal Convolutional Network (TCN).

3.2.1. Skeleton representation and preprocessing

To enhance the input quality, we applied normalization and standardization to the x and z coordinates of the skeleton joints. In addition, we assigned a confidence score to each joint based on its Mean Absolute Error (MAE) from the training set:

$$\text{KeypointScore}_i = 0.85 - 0.1 \cdot \ln \left(\frac{\text{MAE}_i}{\text{MAE}_{\text{avg}}} \right). \quad (6)$$

This score is appended as an extra input dimension to inform the model about the reliability of each joint.

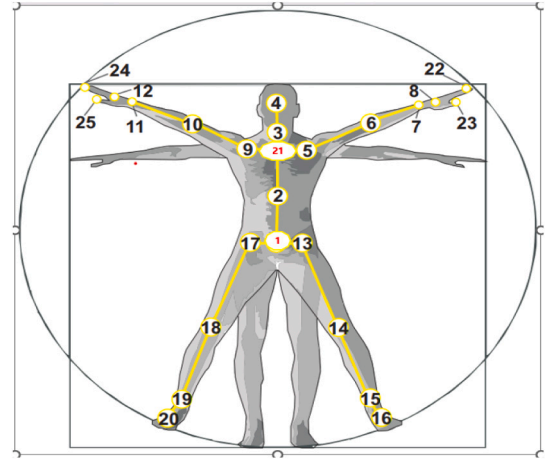


Fig. 3. Skeleton for NTU dataset.

3.2.2. Center point selection

The center joint of the skeleton is used to coordinate normalization and partitioning strategies. As shown in Fig. 3, we selected point 19 (spine shoulder) as the center, which offers better symmetry and consistency.

This stage transforms the estimated skeletons into spatial-temporal graphs that are processed using ST-GCN++ to predict activity classes.

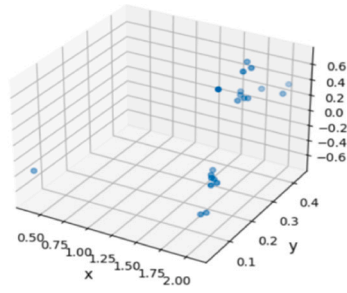
3.2.3. Action classification loss

For the action classification stage, we treated HAR as a multi-class classification problem. Given the estimated skeleton sequence and the corresponding action label, the model learned to minimize the categorical cross-entropy loss. This loss function compares the predicted class probabilities with the ground truth label:

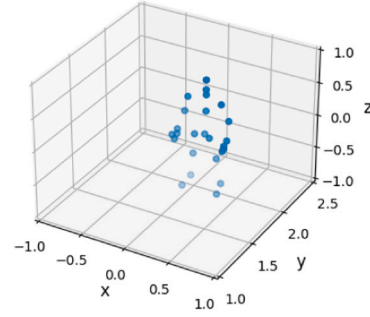
$$\mathcal{L}_{\text{action}} = - \sum_{i=1}^C y_i \log(\hat{y}_i), \quad (7)$$

where C is the number of activity classes, y_i is the one-hot encoded ground truth label for class i , and $\hat{y}_i$ is the predicted probability for class i obtained from the softmax output of the model. The two stages are trained **separately and sequentially**, not jointly. First, SK-DGCNN is trained to completion on the MARS dataset using the shape-preserving loss (Eq. (4)). Once trained, SK-DGCNN generates skeleton predictions for the target datasets (MMActivity and DGHUHA), which are then saved. In the second stage, ST-GCN++ is trained using these pre-generated skeletons with the cross-entropy loss, while SK-DGCNN weights remain frozen. This decoupled training strategy ensures stable optimization, prevents gradient conflicts between pose estimation and classification objectives, and allows independent tuning of each stage.

3D Point Cloud Extracted from mmwave Radar



(a) Original Point Cloud



(b) Estimated Pose by SK-DGCNN

Fig. 4. Input and output of SK-DGCNN.

Table 1

Activity-wise sample distribution in the MMAActivity dataset.

Class	Train samples	Test samples
Boxing	1882	492
Jack	1853	453
Jump	1738	460
Squats	1917	462
Walk	2233	539

Table 2

Activity-wise sample distribution in the DGUHA dataset.

Class	Train samples	Test samples
Running	2350	617
Jumping	2355	587
Sitting down and standing up	2354	601
Both upper limb extension	2341	604
Falling forward	2344	613
Right limb extension	2434	566
Left limb extension	2418	561

4. Experiments

4.1. Datasets

MARS Dataset. The MARS dataset (An & Ogras, 2021) is used to train the SK-DGCNN pose estimation model. It was collected using the Texas Instruments (TI) IWR1443 Boost mmWave radar for point cloud data and the Microsoft Kinect V2 sensor for skeleton data. The dataset includes 10 action classes: left upper limb extension, right upper limb extension, both upper limb extensions, left front lunge, right front lunge, squat, left side lunge, right side lunge, left limb extension, and right limb extension. These synchronized point cloud and skeleton pairs make it suitable for supervised training of pose estimation models.

MMAActivity Dataset. The MMAActivity dataset (Singh et al., 2019) was collected using a TI IWR1443 mmWave radar. It includes five labeled activities as shown in Table 1. Voxelization was originally used to convert the radar signal into a 4D tensor of size $60 \times 10 \times 32 \times 32$, which is computationally intensive. In our approach, we replace voxelization with direct skeleton estimation using SK-DGCNN.

DGUHA Dataset. The DGUHA dataset (Lee & Kim, 2023) includes synchronized point cloud and skeleton data collected using a TI IWR1443BOOST radar and Microsoft Kinect V4 sensor. It involves 19 participants with an average age of 23, performing seven activities as shown in Table 2. Each activity is recorded over approximately 20 s, resulting in a rich and diverse dataset for evaluating HAR performance.

3D Pose Estimation Converting from Point Cloud

4.2. Evaluation metrics

We use two evaluation metrics. Top-1 accuracy measures the percentage of correctly classified samples, while macro F1-score evaluates class-balanced performance. The macro F1-score is computed as:

$$\text{MacroF1} = \frac{1}{C} \sum_{i=1}^C \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}, \quad (8)$$

where C is the total number of activity classes.

Implementation Details. All experiments were conducted on a machine equipped with an Intel Core i9-9900K CPU running at 3.6 GHz, 32 GB of RAM, and an NVIDIA GeForce RTX 2080 Ti GPU, running Ubuntu 20.04 LTS. The implementation was done using Python with PyTorch and TensorFlow frameworks. For the *pose estimation stage* (SK-DGCNN), we used a learning rate of 0.0005 with the Adam optimizer. The model was trained for 150 epochs with a batch size of 16. The number of nearest neighbors (k) in the K-NN graph was set to 20. For the *action classification stage* (ST-GCN++), we used a learning rate of 0.1 with the SGD optimizer and a momentum value of 0.9. The batch size was set to 32, and the model was trained for 72 epochs. A cosine annealing learning rate scheduler was used to stabilize training and encourage convergence. For all reported results, the SK-DGCNN pose estimation weights were frozen during ST-GCN++ training, meaning the skeleton predictions were generated once and then used as fixed input to the classification stage. This two-stage approach ensures computational efficiency and stable training.

Computational Complexity. SK-DGCNN processes radar point clouds with 25 points per frame using 3 EdgeConv blocks (channels: $64 \rightarrow 64 \rightarrow 128$) with ECA attention modules after each block. The model contains approximately 1.94M parameters and requires 40.1 MFLOPs (0.040 GFLOPs) per frame. With a model size of 7.4 MB and total memory footprint of approximately 10 MB (including activations for batch size 16), SK-DGCNN achieved inference times of 1–3 ms per frame on an RTX 2080 Ti GPU, enabling real-time processing at over 300 frames per second. The lightweight architecture of SK-DGCNN makes it highly suitable for deployment in resource-constrained edge devices for real-world HAR applications.

4.3. Results and discussion

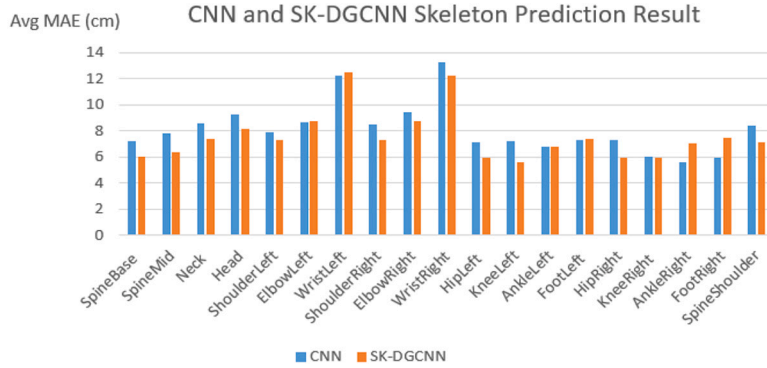
4.3.1. Pose estimation performance

We compare the performance of our proposed SK-DGCNN model against the baseline CNN. Table 3 presents the Mean Absolute Error (MAE) for 19 joints across all three spatial coordinates. The blue values represent the baseline CNN performance, while the red values indicate SK-DGCNN results. On average, SK-DGCNN achieves a lower MAE of 7.58 cm compared to 8.13 cm by CNN.

Table 3

Results of body joints estimation on MMActivity dataset using the simple CNN with provided pre-trained weight in **Blue** and the SK-DGCNN in **Red**.

No	Joint name	X(Horizontal) (cm)		Y(Depth) (cm)		Z(Vertical) (cm)		Average MAE (cm)	
1	Spine Base	7.7608	6.5479	6.3817	4.2005	7.6107	7.2012	7.2511	5.9832
2	Spine Mid	8.4554	7.2735	6.4079	3.6276	8.6467	8.2788	7.8367	6.3933
3	Neck	9.2186	8.2896	6.9313	4.0895	9.6182	9.8244	8.5894	7.4012
4	Head	10.1678	9.0938	7.400	4.4581	10.3114	10.8887	9.2931	8.1469
5	Shoulder Left	9.6090	8.6143	5.3873	4.412	8.6440	8.8038	7.8801	7.2767
6	Elbow Left	10.2758	9.9114	5.8515	5.8585	9.7669	10.4707	8.6314	8.7469
7	Wrist Left	12.8680	13.4087	6.7409	6.6938	17.2048	17.335	12.271	12.4792
8	Shoulder Right	9.1604	8.2727	7.3128	4.628	8.9479	9.026	8.4737	7.3089
9	Elbow Right	9.8240	9.9283	8.7006	6.014	9.8682	10.408	9.4642	8.7834
10	Wrist Right	13.3914	12.9722	8.5864	6.5052	17.7021	17.2424	13.2266	12.24
11	Hip Left	7.6272	6.5797	6.1941	4.243	7.4416	7.0459	7.0876	5.9562
12	Knee Left	8.1499	6.8381	9.9147	5.0213	3.4750	4.9106	7.1798	5.59
13	Ankle Left	8.2115	8.254	9.6362	5.7118	2.4896	6.481	6.7791	6.8156
14	Foot Left	8.5015	8.6928	10.1099	6.2628	3.242	7.1324	7.2845	7.3627
15	Hip Right	7.6212	6.4438	6.6616	4.2715	7.534	7.1335	7.2723	5.9496
16	Knee Right	7.4206	7.2951	6.8102	5.4335	3.8988	5.2127	6.0432	5.9804
17	Ankle Right	7.7786	8.689	5.594	5.9045	3.3748	6.5688	5.5825	7.0541
18	Foot Right	8.2731	9.2493	5.763	5.9407	3.888	7.1602	5.9747	7.4501
19	Spine Shoulder	9.0320	8.0486	6.7375	3.8661	9.3895	9.3947	8.3863	7.1031
Average		9.1235	8.6528	7.2169	5.1128	8.0555	8.9747	8.1320	7.5801

**Fig. 5.** Comparison of the simple CNN and the SK-DGCNN skeleton prediction results on MARS dataset.

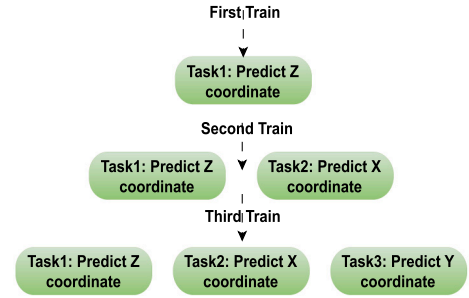
Improvements are especially noticeable for torso joints (e.g., spine base, spine mid, neck), while marginal degradation is observed on limb extremities such as wrists and ankles. These joints are inherently more challenging due to occlusion and motion complexity in radar data. Fig. 4(a) shows the input point cloud, and Fig. 4(b) illustrates qualitative predictions for the corresponding point cloud. Fig. 5 shows a per-joint comparison, indicating that SK-DGCNN generates more coherent and structurally aligned skeletons than the CNN baseline.

4.3.2. Effectiveness of shape-preserving loss

Although MAE is a widely used metric for regression tasks, it fails to capture the global pose structure. To address this, we propose a combined loss function using MSE and KL divergence (Eq. (3)). While MSE penalizes local coordinate deviations, KLDiv encourages the preservation of the overall shape by treating the skeleton as a spatial distribution. The benefit of this loss formulation is indirectly validated by the improved downstream classification accuracy. Despite similar MAE values, skeletons generated using shape-aware loss result in significantly better HAR performance (discussed below), indicating that the model has learned to preserve meaningful spatial structures beyond pointwise accuracy.

4.3.3. Impact of sequential multitask learning

We employed a sequential multitask learning strategy to stabilize training across different coordinate axes. As shown in Fig. 6, the model is first trained on one axis (e.g., z), followed by x and y . This ordering allows the model to converge to a stable representation of joint

**Fig. 6.** Sequential multitask learning in SK-DGCNN.

depth relative to the skeleton center point before modeling lateral and forward-backward motion.

In particular, training on the z axis first followed by x and y was found to be the most effective. Fig. 7 shows reduced validation losses when trained in this sequence. This strategy improves shape consistency and reduces variance across predictions in challenging joints.

4.3.4. Action classification performance

To demonstrate the reproducibility and robustness of our approach, we conducted three independent training runs with different random initializations for each dataset. Tables 4 and 5 present the Top-1 accuracy and F1-score comparisons against previous methods. On MMActivity, our model achieves a mean accuracy of $99.59 \pm 0.22\%$ (best:

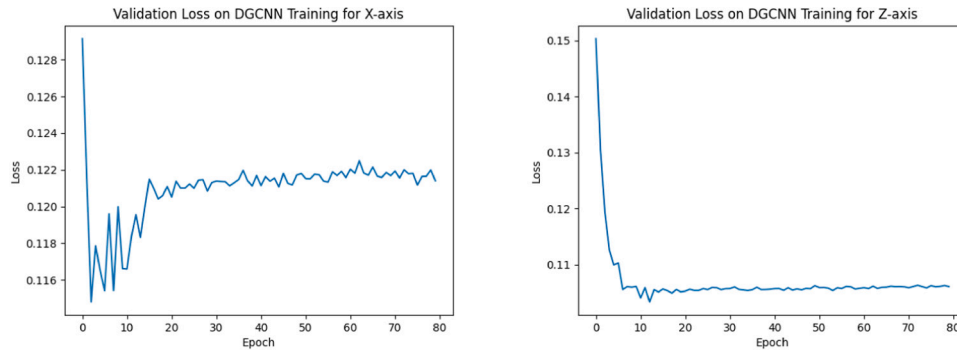


Fig. 7. Validation loss for SK-DGCNN trained on MARS Dataset. (a) X-axis (b) Z-axis.

Table 4

Comparison of action classification performance on the MMActivity dataset.

No.	Classifier	Accuracy (%)	F1-Score (%)
1	SVM (Singh et al., 2019)	63.74	N/A
2	MLP (Singh et al., 2019)	80.34	N/A
3	Bi-directional LSTM (Singh et al., 2019)	88.42	N/A
4	CNN+Bi-LSTM (Singh et al., 2019)	90.47	N/A
5	CNN+Bi-LSTM with Pre-Train Singh et al. (2019)	92.00	N/A
6	ST-GCN with CNN skeletons (Lee & Kim, 2022)	96.55	N/A
7	EdgeActNet (Luo et al., 2023)	97.63	97.44
8	ST-GCN++ with SK-DGCNN (ours)	99.59 ± 0.22	99.59 ± 0.22

Table 5

Comparison of action classification performance on the DGUHA dataset.

No.	Classifier	Accuracy (%)	F1-Score (%)
1	MTGEA (Lee & Kim, 2023)	93.79	N/A
2	EdgeActNet (Luo et al., 2023)	95.03	95.03
3	ST-GCN++ (SK-DGCNN 2D)	94.03	93.89
4	ST-GCN++ (SK-DGCNN 2D with pre-trained x,z)	94.61	94.37
5	ST-GCN++ (SK-DGCNN 2D with pre-trained z,x)	98.72 ± 0.33	98.72 ± 0.33

Table 6

Comparison of the results on the MMActivity dataset to select the best preprocessing method.

No.	Pose estimation	2D/3D	Preprocessing	Top-1 accuracy (%)	F1-Score (%)
1	SK-DGCNN	3D	No	78.83	78.89
2	SK-DGCNN	3D	Normalized	86.95	86.89
3	CNN	3D	Normalized	43.76	43.45
4	SK-DGCNN	2D	Normalized	99.28	99.27
5	SK-DGCNN	2D	Normalized and Standardized	99.73	99.62

99.84%, worst: 99.41%) with $100.00 \pm 0.00\%$ Top-5 accuracy across all runs, surpassing EdgeActNet (Luo et al., 2023), ST-GCN (Lee & Kim, 2022), and CNN-based LSTM methods (Singh et al., 2019). The low standard deviation ($\pm 0.22\%$) demonstrates the stability and our framework, and minimal sensitivity to random initialization.

Similarly, on the DGUHA dataset, as shown in Table 5, our model achieved a mean accuracy of $98.72 \pm 0.33\%$ (best: 99.01%, worst: 98.36%) with $99.96 \pm 0.04\%$ Top-5 accuracy across three runs, again surpassing EdgeActNet and MTGEA. The minimal variance observed across multiple training runs ($\text{std} < 0.35\%$) on both datasets indicates that our model's performance is not dependent on random initialization and suggests genuine generalization rather than overfitting to specific data splits. These results confirm the generalizability and reproducibility of our method across datasets.

We also performed a preprocessing ablation study as presented in Table 6. Results indicate that converting skeletons to 2D and applying both normalization and standardization yields the best classification performance. Models trained with 3D skeletons or without preprocessing underperform, highlighting the importance of data normalization for graph-based models.

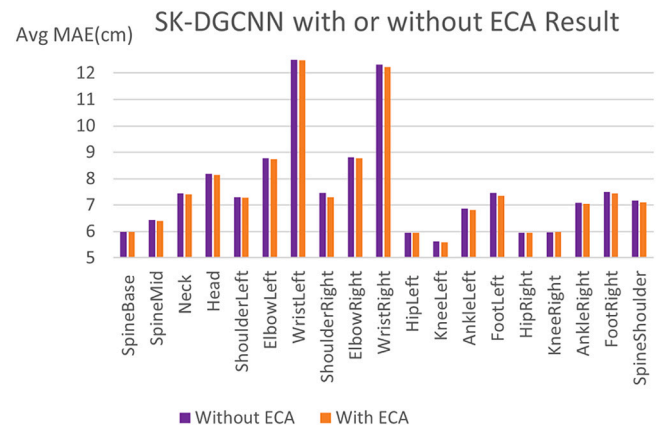


Fig. 8. SK-DGCNN with or without ECA layer result on MARS dataset.

Table 7

Comparison of results with and without our proposed loss function on DGUHA dataset.

Classifier	Top-1 (%)	Top-5 (%)	F1 (%)
MSE	99.16	99.75	99.15
MSE + KLDiv	99.25	100	99.25

4.3.5. Ablation study: Loss function and ECA attention

Table 7 presents an ablation study on the effect of the shape-preserving loss function. The model trained with only MSE achieves a Top-1 accuracy of 99.16%, while the addition of KLDiv increases accuracy to 99.25% and achieves 100% Top-5 accuracy. While the Top-1 improvement of 0.09% may appear numerically small, it is practically significant in this context. First, at accuracy levels above 99%, each percentage point represents a substantial reduction in error from 0.84% to 0.75%. Second, the jump from 99.75% to 100% Top-5 accuracy indicates that the KLDiv term eliminates all cases where the correct class was outside the top prediction, demonstrating more confident and reliable predictions. Third, in safety-critical applications such as fall detection or patient monitoring, even marginal accuracy improvements can prevent false alarms or missed events, directly impacting user trust and system deployment viability. This suggests that including shape information in the loss helps disambiguate visually similar activities by preserving global pose structure. We also evaluate the impact of the Efficient Channel Attention (ECA) layer. As shown in Fig. 8, models with ECA consistently yield lower MAE across joints compared to their non-attentive counterparts. The ECA layer helps emphasize informative features while remaining lightweight, making it suitable for edge deployment.

4.3.6. Per-class performance

To provide a deeper insight into model robustness and address concerns about overfitting or data leakage, we present per-class performance metrics and confusion matrices for both datasets in Fig. 9. On MMActivity, all five activities achieve precision and recall above 98.78%, with F1-scores ranging from 0.9901 (Jack) to 0.9944 (Walk). This demonstrates balanced performance across kinematically diverse actions. The confusion matrix reveals minimal off-diagonal elements, with only 19 total misclassifications out of 2406 test samples. Similarly, on DGUHA, all seven activities achieve precision and recall above 99.35%, with F1-scores between 0.9950 (Falling Forward) and 0.9991 (Both Upper Limb Extension). The normalized confusion matrices show that most classes achieve > 99% correct classification along the diagonal, with negligible confusion between activities. The balanced, high performance across all classes, regardless of sample size or complex activity suggests robust generalization without overfitting to subject-dependent features.

4.4. Limitations and future work

While our framework achieves competitive performance on the evaluated benchmarks, several limitations warrant acknowledgment. The datasets used consist of controlled indoor environments with single-person scenarios performing kinematically distinct activities, which may not reflect more challenging real-world conditions such as multi-person scenes, outdoor settings, or fine-grained activity recognition. Our per-joint analysis reveals that limb extremities (wrists, ankles) exhibit higher estimation errors due to smaller radar cross-sections and motion blur. The current two-stage training approach, while modular and interpretable, prevents end-to-end optimization where classification gradients could guide pose estimation. Future work should explore joint training strategies, extend the framework to multi-person scenarios, incorporate temporal modeling for trajectory smoothing, and evaluate cross-dataset generalization to radars with different specifications. Additionally, deployment on resource-constrained edge devices would benefit from model compression techniques such as quantization and pruning.

Table A.8

SK-DGCNN architecture specification.

Layer	In→Out	Operation	Notes
<i>Input: (16,25,5); k=20</i>			
EdgeConv-1	10→64	Conv2d+BN+LReLU	Edge: $(p_i - p_j, p_i)$
ECA-1	64→64	Ch. attn	k=3
EdgeConv-2	128→64	Conv2d+BN+LReLU	Concat(64×2)
ECA-2	64→64	Ch. attn	k=3
EdgeConv-3	128→128	Conv2d+BN+LReLU	Concat(64×2)
ECA-3	128→128	Ch. attn	k=3
Aggregate	256→256	Concat	(64+64+128)
Global Conv	256→512	Conv1d+BN+LReLU	–
ECA-4	512→512	Ch. attn	k=3
Pooling	512→1024	Max+Avg	Concat
MLP-1	1024→512	Lin+BN+Drop(0.4)+LReLU	–
MLP-2	512→2432	Lin+BN+Drop(0.4)+LReLU	19×128
Reshape	2432→(19,128)	Reshape	–
Out-x	128→1	Linear	x-coord
Out-z	128→1	Linear	z-coord
<i>Output: (batch,19,2)</i>			

5. Conclusion

In this paper, we present a two-stage framework for HAR using radar-based point cloud data. To address the challenges of sparsity and irregularity in radar inputs, we propose SK-DGCNN, a dynamic graph convolutional model that estimates 3D human skeletons without requiring voxelization. By integrating Efficient Channel Attention (ECA) and a shape-preserving loss function that combines MSE and KL divergence, SK-DGCNN captures both joint-level precision and global pose structure. We use the predicted skeletons as input to ST-GCN++, a spatio-temporal graph convolutional network, to classify human activities. The full pipeline achieves strong and reproducible performance on two public datasets, MMActivity ($99.59 \pm 0.22\%$) and DGUHA ($98.72 \pm 0.33\%$) with minimal variance across multiple training runs, demonstrating its effectiveness, efficiency, and applicability to controlled indoor activity recognition scenarios. Our framework offers a privacy-preserving, lightweight solution suitable for deployment in real-world environments such as eldercare, patient monitoring, and indoor surveillance. In future work, we plan to explore end-to-end training of the full pipeline, extend the system to multi-person scenarios, and incorporate multi-sensor data for more robust activity recognition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by NSERC Discovery RGPIN-2018-05550. The source code and pre-trained models will be made publicly available upon acceptance of this manuscript.

Appendix. SK-DGCNN architecture details

This appendix provides complete architectural specifications for SK-DGCNN to ensure reproducibility.

A.1. Network architecture

Table A.8 summarizes the complete SK-DGCNN architecture with layer-by-layer specifications.

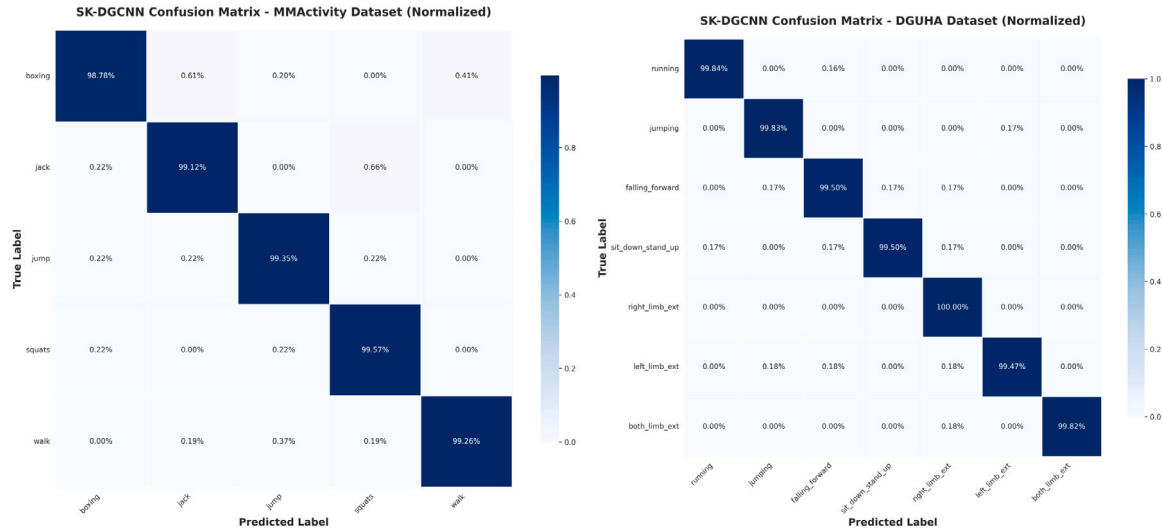


Fig. 9. Normalized confusion matrices showing per-class classification performance (representative run from multi-seed evaluation). (a) MMActivity dataset with 5 activities (2406 test samples, mean accuracy: $99.59 \pm 0.22\%$). (b) DGUHA dataset with 7 activities (4149 test samples, mean accuracy: $98.72 \pm 0.33\%$). Both matrices demonstrate minimal off-diagonal confusion and balanced performance across all classes.

Table A.9

Training configuration.

Parameter	Value	Parameter	Value
Optimizer	Adam	β_1, β_2	0.5, 0.999
Learning rate	0.001	Weight decay	0
LR scheduler	MultiStepLR	Milestones	[15, 20]
Gamma	0.1	Batch size	16
Epochs	150	Loss	MSE + KLDiv
Model Statistics			
Parameters	1.94M	FLOPs/frame	40.1M
Model size	7.39 MB	Memory (batch=16)	10 MB
Inference time	1–3 ms	Hardware	RTX 2080 Ti

Table A.10

ST-GCN++ training configuration and dataset splits.

ST-GCN++ configuration		Dataset splits		
Parameter	Value	Dataset	Train	Test
Optimizer	SGD	MARS	24,066	7984
Learning rate	0.1	MMActivity	9623	2406
Momentum	0.9	DGUHA	16,596	4149
Weight decay	0.0005	(subject-independent splits)		
LR scheduler	Cosine			
Batch size	32			
Epochs	72			
Temporal frames	30			
SK-DGCNN weights	Frozen			

A.2. Training configuration and hyperparameters

Tables A.9 present the complete training setup.

A.3. ST-GCN++ and dataset configuration

Table A.10 presents the hyperparameter of STGCN++ and the Dataset configuration.

Data availability

Data will be made available on request.

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